

Introduction to Machine Learning

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Preface

Machine learning addresses the issue of analyzing, reproducing and predicting various mechanisms and processes observable through experiments and data acquisition. With the impetus of large technological companies in need of leveraging information included in the gigantic datasets that they produced or obtained through user data, with the development of new data acquisition techniques in biology, physics or astronomy, with the improvement of storage capacity and high-performance computing, this field has experienced an explosive growth over the past decades, in terms of scientific production and technological impact.

While it is being recognized in some places as a scientific discipline in itself, machine learning (which has received a few almost synonymic denominations across time, including artificial intelligence, machine intelligence or statistical learning), can also be seen as an interdisciplinary field interfacing techniques from traditional domains such as computer science, applied mathematics, and statistics. From statistics, and more specially nonparametric statistics, it borrows its main formalism, asymptotic results and generalization bounds. It also builds on many classical methods that have been developed for estimation and prediction. From computer science, it involves the construction and implementation of efficient algorithms, programming design and architecture. Finally, machine learning leverages classical methods from linear algebra and functional analysis, as well as from convex and nonlinear optimization, fields within which it had also provided new problems and discoveries. It forms a significant part of the larger field commonly called “data science,” which includes methods for storing, sharing and managing data, the development powerful computer architectures for increasingly demanding algorithms, and, importantly, the definition of ethical limits and processes through which data should be used in the modern world.

This book, which originates from lecture notes of a series of graduate course taught in the Department of Applied Mathematics and Statistics at Johns Hopkins University, adopts a viewpoint (or bias) mainly focused on the mathematical and statistical aspects of the subject. Its goal is to introduce the mathematical foundations and techniques that lead to the development and analysis of many of the algorithms that are used today. It is written with the hope to provide the reader with a deeper

understanding of the algorithms made available to her in multiple machine learning packages and software, and that she will be able to assess their prerequisites and limitations, and to extend them and develop new algorithms. Note that, while adopting a presentation with a strong mathematical flavor, we will still make explicit the details of many important machine learning algorithms.

Unsurprisingly, the book will be more accessible to a reader with some background in mathematics and statistics. It assumes familiarity with basic concepts in linear algebra and matrix analysis, in multivariate calculus and in probability and statistics. We tried to place a limit at the use of measure theoretic tools, that are avoided up to a few exceptions, which are be localized and be accompanied with alternative interpretations allowing for a reading at a more elementary level.

The book starts with an introductory chapter that describes notation used throughout the book and serve at a reminder of basic concepts in calculus, linear algebra and probability. It also introduces some measure theoretic terminology, and can be used as a reading guide for the sections that use these tools. This chapter is followed by two chapters offering background material on matrix analysis and optimization. The latter chapter, which is relatively long, provides necessary references to many algorithms that are used in the book, including stochastic gradient descent, proximal methods, etc.

Chapter 4, which is also introductory, illustrates the bias-variance dilemma in machine learning through the angle of density estimation and motivates chapter 5 in which basic concepts for statistical prediction are provided. Chapter 6 provides an introduction to reproducing kernel theory and Hilbert space techniques that are used in many places, before tackling, with chapters 7 to 11, the description of various algorithms for supervised statistical learning, including linear methods, support vector machines, decision trees, boosting, or neural networks.

Chapter 13, which presents sampling methods and an introduction to the theory of Markov chains, starts a series of chapters on generative models, and associated learning algorithms. Graphical models and described in chapters 14 to 16. Chapter 17 introduces variational methods for models with latent variables, with applications to graphical models in chapter 18. Generative techniques using deep learning are presented in chapter 19.

Chapters 20 to 22 focus on unsupervised learning methods, for clustering, factor analysis and manifold learning. The final chapter of the book is theory-oriented and discusses concentration inequalities and generalization bounds.

Chapter 1

General Notation and Background Material

1.1 Linear algebra

1.1.1 Sets and functions

If A is a set, the set of all subsets of A is denoted $\mathcal{P}(A)$. If A and B are two sets, the notation B^A refers to the set of all functions $f : A \rightarrow B$. In particular, $\mathbb{R}^A$ is the space of real-valued functions, and forms a vector space. When A is finite, this space is finite dimensional and can be identified with $\mathbb{R}^{|A|}$, where $|A|$ denotes the cardinality (number of elements) of A .

The indicator function of a subset C of A will be denoted $\mathbf{1}_C : A \rightarrow \{0, 1\}$, with $\mathbf{1}_C(x) = 1$ if $x \in C$ and 0 otherwise. We will sometimes write $\mathbf{1}_{x \in C}$ for $\mathbf{1}_C(x)$.

1.1.2 Vectors

Elements of the d -dimensional Euclidean space $\mathbb{R}^d$ will be denoted with letters such as x, y, z , and their coordinates will be indexed as parenthesized exponents, so that

$$x = \begin{pmatrix} x^{(1)} \\ \vdots \\ x^{(d)} \end{pmatrix}$$

(we will always identify element of $\mathbb{R}^d$ with column vectors). We will not distinguish in the notation between “points” in $\mathbb{R}^d$, seen as an affine space, and “vectors” in $\mathbb{R}^d$, seen as a vector space. The vectors $\mathbf{0}_d$ and $\mathbf{1}_d$ will denote the d -dimensional vectors with all coordinates equal to 0 and 1, respectively. The identity matrix in $\mathbb{R}^d$ will be denoted $\text{Id}_{\mathbb{R}^d}$. The canonical basis of $\mathbb{R}^d$, provided by the columns of $\text{Id}_{\mathbb{R}^d}$ will be denoted $\mathbf{e}_1, \dots, \mathbf{e}_d$.

The Euclidean norm of a vector $x \in \mathbb{R}^d$ is denoted $|x|$ with

$$|x| = \left((x^{(1)})^2 + \dots + (x^{(d)})^2 \right)^{1/2}.$$

It will sometimes be denoted $|x|_2$, identifying it as a member of the family of ℓ^p norms

$$|x|_p = \left((x^{(1)})^p + \dots + (x^{(d)})^p \right)^{1/p} \quad (1.1)$$

for $p \geq 1$. One can also define $|x|_p$ for $0 < p < 1$, using (1.1), but in this case one does not get a norm because the triangle inequality $|x + y|_p \leq |x|_p + |y|_p$ is not true in general. The family is interesting, however, because it approximates, in the limit $p \rightarrow 0$, the number of non-zero components of x , denoted $|x|_0$, which is a measure of sparsity. Note that we also use the notation $|A|$ to denote the cardinality (number of elements) of a set A , hopefully without risk of confusion.

While we use single bars ($|x|$) to represent norms of finite-dimensional vectors, we will use double bars ($\|h\|$) for infinite-dimensional objects.

1.1.3 Matrices

The set of $m \times d$ real matrices with real entries is denoted $\mathcal{M}_{m,d}(\mathbb{R})$, or simply $\mathcal{M}_{m,d}$ ($\mathcal{M}_{d,d}$ will also be denoted $\mathcal{M}_d$). The set of invertible $d \times d$ matrices will be denoted $\mathcal{GL}_d(\mathbb{R})$.

Given m column vectors $x_1, \dots, x_m \in \mathbb{R}^d$, the notation $[x_1, \dots, x_m]$ refers to the d by m matrix with j^{th} column equal to x_j , so that, for example, $\text{Id}_{\mathbb{R}^d} = [\epsilon_1, \dots, \epsilon_d]$.

Entry (i, j) in a matrix $A \in \mathcal{M}_{m,d}(\mathbb{R})$ will either be denoted $A(i, j)$ or $A_j^{(i)}$. The rows of A will be denoted $A^{(1)}, \dots, A^{(m)}$ and the columns $A_1, \dots, A_m$.

The operator norm of a matrix $A \in \mathcal{M}_{m,d}$ is defined by

$$|A|_{\text{op}} = \max\{|Ax| : x \in \mathbb{R}^d, |x| = 1\}.$$

The space of $d \times d$ real symmetric matrices is denoted $\mathcal{S}_d$, and its subsets containing positive semi-definite (resp. positive definite) matrices is denoted $\mathcal{S}_d^+$ (resp. $\mathcal{S}_d^{++}$). If $m \leq d$, $\mathcal{O}_{m,d}$ denotes the set of $m \times d$ matrices A such that $AA^T = \text{Id}_{\mathbb{R}^m}$, and one writes $\mathcal{O}_d$ for $\mathcal{O}_{d,d}$, the space of d -dimensional orthogonal matrices. Finally, $\mathcal{SO}_d$ is the subset $\mathcal{O}_d$ containing orthogonal matrices with determinant 1, i.e., rotation matrices.

1.1.4 Multilinear maps

A k -linear map is a function $a : (x_1, \dots, x_k) \mapsto a(x_1, \dots, x_k)$ defined on $(\mathbb{R}^d)^k$ with values in $\mathbb{R}^q$ which is linear in each of its variables. The mapping is symmetric if its value is unchanged after any permutation of the variables. If $k = 2$ and $q = 1$, one also says that a is a bilinear form. The norm of a k -linear map is defined as

$$|a| = \max\{a(x_1, \dots, x_k) : |x_j| \leq 1, j = 1, \dots, k\}$$

so that

$$|a(x_1, \dots, x_k)| \leq |a| \prod_{j=1}^k |x_j|$$

for all $x_1, \dots, x_k \in \mathbb{R}^d$.

A symmetric bilinear (i.e., 2-linear) form a is called positive semidefinite if $a(x, x) \geq 0$ for all $x \in \mathbb{R}^d$, and positive definite if it is positive semi-definite and $a(x, x) = 0$ if and only if $x = 0$. Symmetric bilinear forms can always be expressed in the form $a(x, y) = x^T A y$ for some symmetric matrix A , and a is positive (semi-)definite if and only if A is also. Analogous statements hold for negative (semi-)definite forms and matrices. We will use the notation $A > 0$ (resp. ≥ 0) to indicate that A is positive definite (resp. positive semidefinite). Note that, if $a(x, y) = x^T A y$ for $A \in \mathcal{S}_d$, then $|a| = |A|_{\text{op}}$.

1.2 Topology

1.2.1 Open and closed sets in $\mathbb{R}^d$

The open balls in $\mathbb{R}^d$ will be denoted

$$B(x, r) = \{y \in \mathbb{R}^d : |y - x| < r\},$$

with $x \in \mathbb{R}^d$ and $r > 0$. The closed balls are denoted $\bar{B}(x, r)$ and contain all y 's such that $|y - x| \leq r$. A set $U \subset \mathbb{R}^d$ is open if and only if for any $x \in U$, there exists $r > 0$ such that $B(x, r) \subset U$. A set $\Gamma \subset \mathbb{R}^d$ is closed if its complement, denoted

$$\Gamma^c = \{x \in \mathbb{R}^d : x \notin \Gamma\}$$

is open. The topological interior of a set $A \subset \mathbb{R}^d$ is the largest open set included in A . It will be denoted either by $\mathring{A}$ or $\text{int}(A)$. A point x belongs to $\mathring{A}$ if and only if $B(x, r) \subset A$ for some $r > 0$.

The closure of A is the smallest closed set that contains A and will be denoted either $\bar{A}$ or $\text{cl}(A)$. A point x belongs to $\bar{A}$ if and only if $B(x, r) \cap A \neq \emptyset$ for all $r > 0$. Alternatively, x belongs to $\bar{A}$ if and only if there exists a sequence (x_k) that converges to x with $x_k \in A$ for all k .

1.2.2 Compact sets

A compact set in $\mathbb{R}^d$ is a set Γ such that any sequence of points in Γ contains a subsequence that converges to some point in Γ . An alternate definition is that, whenever Γ is covered by a collection of open sets, there exists a finite subcollection that still covers Γ .

One can show that compact subsets of $\mathbb{R}^d$ are exactly its bounded and closed subsets.

1.2.3 Metric spaces

A metric space is a space $\mathcal{B}$ equipped with a distance, i.e., a function $\rho : \mathcal{B} \times \mathcal{B} \rightarrow [0, +\infty)$ that satisfies the following three properties.

$$\forall x, y \in \mathcal{B} : \rho(x, y) = 0 \Leftrightarrow x = y, \quad (1.2a)$$

$$\forall x, y \in \mathcal{B} : \rho(x, y) = \rho(y, x), \quad (1.2b)$$

$$\forall x, y, z \in \mathcal{B} : \rho(x, z) \leq \rho(x, y) + \rho(y, z). \quad (1.2c)$$

Equation (1.2c) is called the triangle inequality. The norm of the difference between two points: $\rho(x, y) = |x - y|$, is a distance on $\mathbb{R}^d$. The definition of open and closed subsets in metric spaces is the same as above, with $\rho(x, y)$ replacing $|x - y|$, and one says that (x_n) converges to x if and only if $\rho(x_n, x) \rightarrow 0$.

Compact subsets are also defined in the same way, but are not necessarily characterized as bounded and closed.

1.3 Calculus

1.3.1 Differentials

If $x, y \in \mathbb{R}^d$, we will denote by $[x, y]$ the closed segment delimited by x and y , i.e., the set of all points $(1 - t)x + ty$ for $0 \leq t \leq 1$. One denotes by $[x, y)$, $(x, y]$ and (x, y) the semi-open or open segments, with appropriate strict inequality for t . (Similarly to the notation for open intervals, whether (x, y) denotes an open segment or a pair of points will always be clear from the context.)

The derivative of a differentiable function $f : t \mapsto f(t)$ from an interval $I \subset \mathbb{R}$ to $\mathbb{R}$ will be denoted by ∂f , or $\partial_t f$ if the variable t is well identified. Its value at $t_0 \in I$ is denoted either as $\partial f(t_0)$ or $\partial f|_{t=t_0}$. Higher derivatives are denoted as $\partial^k f$, $k \geq 0$, with the usual convention $\partial^0 f = f$. Note that notation such as $f', f'', f^{(3)}$ will *never* refer to derivatives.

In the following, U is an open subset of $\mathbb{R}^d$. If f is a function from U to $\mathbb{R}^m$, we let $f^{(i)}$ denote the i^{th} component of f , so that

$$f(x) = \begin{pmatrix} f^{(1)}(x) \\ \vdots \\ f^{(m)}(x) \end{pmatrix}$$

for $x \in U$. If $d = 1$, and f is differentiable, the derivative of f at x is the column vector of the derivatives of its components,

$$\partial f(x) = \begin{pmatrix} \partial f^{(1)}(x) \\ \vdots \\ \partial f^{(m)}(x) \end{pmatrix}$$

For $d \geq 1$ and $j \in \{1, \dots, d\}$, the j^{th} partial derivative of f at x is

$$\partial_j f(x) = \partial(t \mapsto f(x + t\epsilon_j))|_{t=0} \in \mathbb{R}^m,$$

where $\epsilon_1, \dots, \epsilon_d$ form the canonical basis of $\mathbb{R}^d$. If the notation for the variables on which f depends is well understood from the context, we will alternatively use $\partial_{x_j} f$. (For example, if $f : (\alpha, \beta) \mapsto f(\alpha, \beta)$, we will prefer $\partial_\alpha f$ to $\partial_1 f$.) The differential of f at x is the linear mapping from $\mathbb{R}^d$ to $\mathbb{R}^m$ represented by the matrix

$$df(x) = [\partial_1 f(x), \dots, \partial_d f(x)].$$

It is defined so that, for all $h \in \mathbb{R}^d$

$$df(x)h = \partial(t \mapsto f(x + th))|_{t=0}$$

where the right-hand side is the directional derivative of f at x in the direction h . Note that, if $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (i.e., $m = 1$), $df(x)$ is a row vector. If f is differentiable on U and $df(x)$ is continuous as a function of x , one says that f is continuously differentiable, or C^1 .

Differentials obey the product rule and the chain rule. If $f, g : U \rightarrow \mathbb{R}$, then

$$d(fg)(x) = f(x)dg(x) + g(x)df(x).$$

If $f : U \rightarrow \mathbb{R}^m$, $g : \tilde{U} \subset \mathbb{R}^k \rightarrow U$, then

$$d(f \circ g)(x) = df(g(x))dg(x).$$

If $d = m$ (so that $df(x)$ is a square matrix), we let $\nabla \cdot f(x) = \text{trace}(df(x))$, the divergence of f .

The terms “derivative” and “differential” are mostly interchangeable, although one often uses “derivative” for differentiation with respect to a scalar variable.

The Euclidean gradient of a differentiable function $f : U \rightarrow \mathbb{R}$ is $\nabla f(x) = df(x)^T$. More generally, one defines the gradient of f with respect to a tensor field $x \mapsto A(x)$ taking values in $\mathcal{S}_d^{++}$, as the vector $\nabla_A f(x)$ that satisfies the relation

$$df(x)h = \nabla_A f(x)^T A(x)h$$

for all $h \in \mathbb{R}^d$, so that

$$\nabla_A f(x) = A(x)^{-1} df(x)^T. \quad (1.3)$$

In particular, the Euclidean gradient is associated with $A(x) = \text{Id}_{\mathbb{R}^d}$ for all x . With some abuse of notation, we will denote $\nabla_A f = A^{-1} \nabla f$ when A is a fixed matrix, therefore identified with the constant tensor field $x \mapsto A$.

1.3.2 Important examples

We here compute, as an illustration and because they will be useful later, the differential of the determinant and the inversion in matrix spaces.

Recall that, if $A = [a_1, \dots, a_d] \in \mathcal{M}_d$ is a d by d matrix, with $a_1, \dots, a_d \in \mathbb{R}^d$, $\det(A)$ is a d -linear form $\delta(a_1, \dots, a_d)$ which vanishes when two columns coincide and such that $\delta(e_1, \dots, e_d) = 1$. In particular δ changes signs when two of its columns are inverted. It follows from this that

$$\begin{aligned} \partial_{a_{ij}} \det(A) &= \delta(a_1, \dots, a_{i-1}, e_j, a_{j+1}, \dots, a_d) \\ &= (-1)^{i-1} \delta(e_j, a_1, \dots, a_{i-1}, \dots, a_d) = (-1)^{i+j} \det A^{(ij)}, \end{aligned}$$

where $A^{(ij)}$ is the matrix A with row i and column j removed. We therefore find that the differential of $A \mapsto \det(A)$ is the mapping

$$H \mapsto \text{trace}(\text{cof}(A)^T H) \quad (1.4)$$

where $\text{cof}(A)$ is the matrix composed of co-factors $(-1)^{i+j} \det A^{(ij)}$. As a consequence, if A is invertible, then the differential of $\log |\det(A)|$ is the mapping

$$H \mapsto \text{trace}(\det(A)^{-1} \text{cof}(A)^T H) = \text{trace}(A^{-1} H) \quad (1.5)$$

Consider now the function $I(A) = A \mapsto A^{-1}$ defined on $\mathcal{GL}_d(\mathbb{R})$, which is an open subset of $\mathcal{M}_d(\mathbb{R})$. Using $AI(A) = \text{Id}_{\mathbb{R}^d}$ and the product rule, we get

$$A(dI(A)H) + HI(A) = 0$$

or

$$dI(A)H = -A^{-1}HA^{-1}. \quad (1.6)$$

1.3.3 Higher order derivatives

Higher-order partial derivatives $\partial_{i_k} \cdots \partial_{i_1} f : U \rightarrow \mathbb{R}^m$ are defined by iterating the definition of first-order derivatives, namely

$$\partial_{i_k} \cdots \partial_{i_1} f(x) = \partial_{i_k} (\partial_{i_{k-1}} \cdots \partial_{i_1} f)(x)$$

If all order k partial derivatives of f exist and are continuous, one says that f is k -times continuously differentiable, or C^k and, when true, the order in which the derivatives are taken does not matter. In this case, one typically groups derivatives with the same order using a power notation, writing, for example

$$\partial_1 \partial_2 \partial_1 f = \partial_1^2 \partial_2 f$$

for a C^3 function.

If f is C^k , its k^{th} differential at x is a symmetric k -multilinear map that can also be iteratively defined by (for $h_1, \dots, h_k \in \mathbb{R}^d$)

$$d^k f(x)(h_1, \dots, h_k) = d(d^{k-1} f(x)(h_1, \dots, h_{k-1}))h_k \in \mathbb{R}^m.$$

It is related to partial derivatives through the relation:

$$d^k f(x)(h_1, \dots, h_k) = \sum_{i_1, \dots, i_k=1}^d h_1^{(i_1)} \cdots h_k^{(i_k)} \partial_{i_k} \cdots \partial_{i_1} f(x).$$

When $m = 1$ and $k = 2$, one denotes by $\nabla^2 f(x) = (\partial_i \partial_j f(x), i, j = 1, \dots, n)$ the symmetric matrix formed by partial derivatives of order 2 of f at x . It is called the *Hessian* of f at x and satisfies

$$h_1^T \nabla^2 f(x) h_2 = d^2 f(x)(h_1, h_2).$$

The Laplacian of f is the trace of $\nabla^2 f$ and denoted Δf .

1.3.4 Taylor's theorem

Taylor's theorem, in its integral form, generalizes the fundamental theorem of calculus to higher derivatives. It expresses the fact that, if f is C^k on U and $x, y \in U$ are such that the closed segment $[x, y]$ is included in U , then, letting $h = y - x$:

$$\begin{aligned} f(x+h) = f(x) + df(x)h + \frac{1}{2}d^2 f(x)(h, h) + \cdots + \frac{1}{(k-1)!}d^{k-1} f(x)(h, \dots, h) \\ + \frac{1}{(k-1)!} \int_0^1 (1-t)^{k-1} d^k f(x+th)(h, \dots, h) dt \quad (1.7) \end{aligned}$$

The last term (remainder) can also be written as

$$\frac{1}{k!} \frac{\int_0^1 (1-t)^{k-1} d^k f(x+th)(h, \dots, h) dt}{\int_0^1 (1-t)^{k-1} dt}.$$

If f takes scalar values, then $d^k f(x+th)(h, \dots, h)$ is real and the intermediate value theorem implies that there exists some z in $[x, y]$ such that

$$f(x+h) = f(x) + df(x)h + \frac{1}{2}d^2 f(x)(h, h) + \dots + \frac{1}{(k-1)!}d^{k-1} f(x)(h, \dots, h) + \frac{1}{k!}d^k f(z)(h, \dots, h). \quad (1.8)$$

This is not true if f takes vector values. However, for any M such that $|d^k f(z)| \leq M$ for $z \in [x, y]$ (such M 's always exist because f is C^k), one has

$$\frac{1}{(k-1)!} \int_0^1 (1-t)^{k-1} d^k f(x+th)(h, \dots, h) dt \leq \frac{M}{k!} |h|^k.$$

Equation (1.7) can be written as

$$f(x+h) = f(x) + df(x)h + \frac{1}{2}d^2 f(x)(h, h) + \dots + \frac{1}{k!}d^k f(x)(h, \dots, h) + \frac{1}{(k-1)!} \int_0^1 (1-t)^{k-1} (d^k f(x+th)(h, \dots, h) - d^k f(x)(h, \dots, h)) dt. \quad (1.9)$$

Let

$$\epsilon_x(r) = \max \{ |d^k f(x+h) - d^k f(x)| : |h| \leq r \}.$$

Since $d^k f$ is continuous, $\epsilon_x(r)$ tends to 0 when $r \rightarrow 0$ and we have

$$\int_0^1 (1-t)^{k-1} |d^k f(x+th)(h, \dots, h) - d^k f(x)(h, \dots, h)| dt \leq \frac{|h|^k}{k} \epsilon_x(|h|).$$

This shows that (1.7) implies that

$$f(x+h) = f(x) + df(x)h + \frac{1}{2}d^2 f(x)(h, h) + \dots + \frac{1}{k!}d^k f(x)(h, \dots, h) + \frac{|h|^k}{k!} \epsilon_x(|h|) \quad (1.10)$$

$$= f(x) + df(x)h + \frac{1}{2}d^2 f(x)(h, h) + \dots + \frac{1}{k!}d^k f(x)(h, \dots, h) + o(|h|^k) \quad (1.11)$$

1.4 Probability theory

1.4.1 General assumptions and notation

When discussing probabilistic concepts, we will make the convenient assumption that all random variables are defined on a fixed probability space $(\Omega, \mathbb{P})$. This means that Ω is large enough to include enough randomness to generate all required variables (and implicitly enlarged when needed).

We assume that the reader is familiar with concepts related to discrete random variables (which take values in a discrete or countable space) and their probability mass function (p.m.f.) or continuous variables (with values in $\mathbb{R}^d$ for some d) and their probability density functions (p.d.f.) when they exist. In particular, $X : \Omega \rightarrow \mathbb{R}^d$ is a random variable with p.d.f. f if and only if the expectation of $\varphi(X)$ is given by

$$\mathbb{E}(\varphi(X)) = \int_{\mathbb{R}^d} \varphi(x) f(x) dx$$

for all bounded and continuous functions $\varphi : \mathbb{R}^d \rightarrow [0, +\infty)$. Not all random variables of interest can be categorized as discrete or continuous with a p.d.f., however, and the others are more conveniently handled using measure-theoretic notation as introduced below.

With a few exceptions, we will use capital letters for random variables and small letters for scalars and vectors that represent realizations of these variables. One of these exceptions will be our notation for training data, defined as an independent and identically distributed (i.i.d.) sample of a given random variable. A realization of such a sample will always be denoted $T = (x_1, \dots, x_N)$, which is therefore a series of observations. We will use the notation $\mathbb{T} = (X_1, \dots, X_N)$ for the collection of i.i.d. random variables that generate the training set, so that $T = (X_1(\omega), \dots, X_N(\omega)) = \mathbb{T}(\omega)$ for some $\omega \in \Omega$. Another exception will apply to variables denoted using Greek letters, for which we will use boldface fonts (such as $\alpha, \beta, \dots$).

For a random variable X , the notation $[X = x]$, or $[X \in A]$ refers to subsets of Ω , for example,

$$[X = x] = \{\omega \in \Omega : X(\omega) = x\}.$$

1.4.2 Conditional probabilities and expectation

If $X : \Omega \rightarrow \mathcal{R}_X$ and $Y : \Omega \rightarrow \mathcal{R}_Y$ are discrete random variables, then

$$\mathbb{P}(Y = y \mid X = x) = \mathbb{P}(Y = y, X = x) / \mathbb{P}(X = x)$$

if $\mathbb{P}(X = x) > 0$ and is undefined otherwise. Then, if Y is scalar- or vector-valued and discrete, one defines the conditional expectation of Y given X , denoted $\mathbb{E}(Y | X)$, by

$$\mathbb{E}(Y | X)(\omega) = \sum_{y \in \mathcal{R}_Y} y \mathbb{P}(Y = y | X = X(\omega))$$

for all ω such that $\mathbb{P}(X = X(\omega)) > 0$. Note that $\mathbb{E}(Y | X)$ is a random variable, defined over Ω . It however only depends on the values of X , in the sense that $\mathbb{E}(Y | X)(\omega) = \mathbb{E}(Y | X)(\omega')$ if $X(\omega) = X(\omega')$. We will use the notation

$$\mathbb{E}(Y | X = x) = \sum_{y \in \mathcal{R}_Y} y \mathbb{P}(Y = y | X = x),$$

which is now a function defined on $\mathcal{R}_X$, satisfying $\mathbb{E}(Y | X)(\omega) = \mathbb{E}(Y | X = X(\omega))$.

If X and Y are scalar- or vector-valued and their joint distribution have a p.d.f. $\varphi_{X,Y}$, one defines similarly the conditional p.d.f. of Y given X by

$$\varphi_Y(y | X = x) = \frac{\varphi_{X,Y}(y, x)}{\int_{\mathcal{R}_Y} \varphi_{X,Y}(y', x) dy'}$$

provided that the denominator does not vanish. We will also use the notation $\varphi_Y(y | X)(\omega) = \varphi_Y(y | X = X(\omega))$ for $\omega \in \Omega$. One then defines

$$E(Y | X)(\omega) = \int_{\mathcal{R}_Y} y \varphi_Y(y | X = X(\omega)) dy.$$

In both cases considered above, it is easily checked that the conditional expectation satisfies the properties

(CE1) $\mathbb{E}(Y | X)(\omega)$ only depends on $X(\omega)$

(CE2) For all functions $f : \mathcal{R}_X \rightarrow [0, +\infty)$ (continuous in the case of continuous random variables), one has $\mathbb{E}(\mathbb{E}(Y | X)f(X)) = \mathbb{E}(Yf(X))$.

The proof that our definition of $\mathbb{E}(Y | X)$ for discrete random variables is the only one satisfying these properties is left to the reader. For continuous random variables, assume that a function $g : \mathcal{R}_X \rightarrow \mathcal{R}_Y$ satisfies

$$\mathbb{E}(g(X)f(X)) = \mathbb{E}(Yf(X))$$

for all continuous $f \geq 0$. Then, letting $\varphi_X(x) = \int_{\mathcal{R}_Y} \varphi_{X,Y}(x, y) dy$, which is the marginal p.d.f. of X ,

$$\int_{\mathcal{R}_X} g(x)f(x)\varphi_X(x)dx = \int_{\mathcal{R}_X \times \mathcal{R}_Y} yf(x)\varphi_{X,Y}(x, y)dx dy = \int_{\mathcal{R}_X} \left(\int_{\mathcal{R}_Y} y\varphi_{X,Y}(x, y)dy \right) dx.$$

If we assume that $\varphi_{X,Y}$ is continuous, then this identity being true for all f implies that

$$g(x)\varphi_X(x) = \int_{\mathcal{R}_Y} y\varphi_{X,Y}(x,y)dy$$

so that g is the conditional expectation. If $\varphi_{X,Y}$ is not continuous, then the identity holds everywhere except on an exceptional “negligible” set (see the measure theoretic introduction below). Properties (CE1) and (CE2) provide the definition of the conditional expectation for general random variables.

Taking $g(x) = 1$ for all $x \in \mathcal{R}_X$ in (CE2) yields the well-known identity

$$\mathbb{E}(\mathbb{E}(Y | X)) = \mathbb{E}(Y).$$

Moreover, for any function g defined on $\mathcal{R}_X$ we have

$$\mathbb{E}(Yg(X) | X) = g(X)\mathbb{E}(Y | X),$$

which can be checked by proving that the right-hand side satisfies conditions (i) and (ii).

1.4.3 Measure theoretic probability

As much as possible—but not always—we will avoid relying on measure theory in our discussions, at the expense of sometimes making not fully rigorous or incomplete statements (that readers familiar with this theory will easily complete). However, there will be situations in which the flexibility of the measure-theoretic formalism is needed for the exposition. The following notions may help the reader navigate through these situations (basic references in measure theory are Rudin [176], Dudley [68], Billingsley [31]).

A measurable space is a pair $(S, \mathcal{S})$ where S is a set and $\mathcal{S} \subset \mathcal{P}(S)$ contains S , is stable by complementation (if $A \in \mathcal{S}$, then $A^c = S \setminus A \in \mathcal{S}$), by countable unions and intersections. Such an $\mathcal{S}$ is called a σ -algebra and elements of $\mathcal{S}$ form the measurable subsets of S (relative to the σ -algebra).

A (positive) measure μ on $(S, \mathcal{S})$ is a mapping from $\mathcal{S} \rightarrow [0, +\infty)$ that associates to $A \in \mathcal{S}$ its measure $\mu(A)$, such that the measure of a countable union of disjoint sets is the countable sum of their measures. A function $f : \Omega \rightarrow \mathbb{R}^d$ is called measurable if the inverse images by f of open subsets of $\mathbb{R}^d$ are measurable. More generally, if $(S, \mathcal{S})$ and $(S', \mathcal{S}')$ are measurable spaces, $f : S \rightarrow S'$ is measurable if $f^{-1}(A') \in \mathcal{S}$ for all $A' \in \mathcal{S}'$.

Remark 1.1. In these notes, measurable spaces S will always (and without mention) be a complete metric (or “metrizable”) space with a dense countable subset

(also called a Polish space), and $\mathcal{S}$ the smallest σ -algebra containing all open subsets of S , which is called the Borel σ -algebra.

Also without mention, all measures will be assumed to be “ σ -finite”, which means that there exists a sequence of measurable sets with finite measure whose union is the whole set S . $\blacklozenge$

If μ is a measure, a measurable set A is μ -negligible (or negligible for μ) if there exists $B \in \mathcal{S}$ such that $A \subset B$ and $\mu(B) = 0$ and events are said to happen almost everywhere if their complements are negligible. A countable union of negligible sets is negligible, but this is not true for non countable unions, which may not even be measurable. It is convenient, and always possible, to extend a σ -algebra so that it contains all μ -negligible sets.

The integral of a function $f : S \rightarrow \mathbb{R}^d$ with respect to a measure μ is denoted $\int_S f d\mu$ or $\int_S f(x)\mu(dx)$. This integral is defined, using a limit argument, as a function which is linear in f and such that, for all $A \in \mathcal{S}$,

$$\int_A \mu(dx) = \int_S \mathbf{1}_A(x)\mu(dx) = \mu(A).$$

More precisely, this uniquely defines the integral of linear combinations of indicator functions with finite measures (called “simple functions”), and one then defines $\int_S f d\mu$ for $f : S \rightarrow [0, +\infty)$ as the supremum of the integrals among all simple functions that are no larger than f . After showing that the result is well defined and linear in f , one defines the integral of $f : S \rightarrow \mathbb{R}$ as the difference between those of $\max(f, 0)$ and $\max(-f, 0)$, which is well define as soon as $\int_S |f| d\mu < \infty$, in which case one says that f is μ -integrable.

The Lebesgue measure, $\mathcal{L}_d$, on $\mathbb{R}^d$ provides an important example. For this measure, $\mathcal{S}$ is the σ -algebra generated by open subsets of $\mathbb{R}^d$ (the smallest one that contains all open subsets), and $\int_{\mathbb{R}^d} f(x)\mathcal{L}_d(dx)$ extends the Riemann integral, justifying the alternative notation $\int_{\mathbb{R}^d} f(x)dx$ that we will preferably use. Another important example, especially when S is finite or countable, is the counting measure, denoted $card$, that returns the number of elements of a set, so that $card(A) = |A|$. If S is finite or countable, one generally takes $\mathcal{S} = \mathcal{P}(S)$ (every subset of S is measurable) and the integral is simply the sum:

$$\int_S f(x) card(dx) = \sum_{x \in \mathcal{F}} f(x).$$

1.4.4 Product of measures

Let μ_1 is a measure on $(S_1, \mathcal{S}_1)$ and μ_2 a measure on $(S_2, \mathcal{S}_2)$, in which we assume the situation described in remark 1.1. The “tensor product” of μ_1 and μ_2 is denoted $\mu_1 \otimes \mu_2$. It is a measure on $S_1 \times S_2$ defined by $\mu_1 \otimes \mu_2(A_1 \times A_2) = \mu_1(A_1)\mu_2(A_2)$ for $A_1 \in \mathcal{S}_1$ and $A_2 \in \mathcal{S}_2$ (the σ -algebra on $S_1 \times S_2$ is the smallest one that contains all sets $A_1 \times A_2$, $A_1 \in \mathcal{S}_1$, $A_2 \in \mathcal{S}_2$).

The integral, with respect to the product measure, of a function $f : S_1 \times S_2 \rightarrow \mathbb{R}^d$ is denoted

$$\int_{S_1 \times S_2} f(x_1, x_2) \mu_1(dx_1) \mu_2(dx_2)$$

(rather than

$$\int_{S_1 \times S_2} f(x_1, x_2) \mu_1 \otimes \mu_2(dx_1, dx_2)).$$

Fubini’s theorem justifies integration by parts: if f is integrable with respect to the product measure, then (integrating first in the second variable)

$$F(x_1) = \int_{S_2} f(x_1, x_2) \mu_2(dx_2)$$

is well defined and

$$\int_{S_1 \times S_2} f(x_1, x_2) d(\mu_1 \otimes \mu_2) = \int_{S_1} F(x_1) d\mu_1.$$

(And one has a symmetric statement by integrating first in the first variable.)

The tensor product between more than two measures is defined similarly, with notation

$$\mu_1 \otimes \cdots \otimes \mu_n = \bigotimes_{k=1}^n \mu_k.$$

1.4.5 Relative absolute continuity and densities

If μ and ν are measures on $(S, \mathcal{S})$, one says that ν is absolutely continuous with respect to μ and write $\nu \ll \mu$ if,

$$\forall A \in \mathcal{S} : \mu(A) = 0 \Rightarrow \nu(A) = 0. \quad (1.12)$$

Assume that μ is σ -finite (remark 1.1). The Radon-Nikodym theorem states that $\nu \ll \mu$ with $\nu(S) < \infty$ if and only if ν has a *density* with respect to μ , i.e., there exists a μ -integrable function $\varphi : S \rightarrow [0, +\infty)$ such that

$$\int_S f(x) \nu(dx) = \int_S f(x) \varphi(x) \mu(dx)$$

for all measurable $f : S \rightarrow [0, +\infty)$.

1.4.6 Measure-theoretic probability

When using measure-theoretic probability, we will therefore assume that the pair $(\Omega, \mathbb{P})$ is completed to a triple $(\Omega, \mathcal{A}, \mathbb{P})$ where $\mathcal{A}$ is a σ -algebra and $\mathbb{P}$ a probability measure, that is a (positive) measure on $(\Omega, \mathcal{A})$ such that $\mathbb{P}(\Omega) = 1$. This triple is called a *probability space*. For probability spaces, measurable sets are also called “events” and events that happen with probability one are said to happen “almost surely.”

A random variable X must then also take values in a measurable space, say $(\mathcal{R}_X, \mathcal{S}_X)$, and must be such that, for all $C \in \mathcal{S}_X$, the set $[X \in C]$ belongs to $\mathcal{S}_X$. This justifies the computation of $\mathbb{P}(X \in C)$, which is also denoted $P_X(C)$.

A random variable X taking values in $\mathbb{R}^d$ has a p.d.f. if and only if $P_X \ll \mathcal{L}_d$ and the p.d.f. is the density provided by the Radon-Nikodym theorem. For a discrete random variable (i.e., taking values in a finite or countable set), the p.m.f. of X is also the density of P_X with respect to the counting measure *card* (every discrete measure is absolutely continuous with respect to *card*).

If X is a random variable with values in $\mathbb{R}^d$, the integral of X with respect to $\mathbb{P}$ is the expectation of X , denoted $\mathbb{E}(X)$. More generally, if $(S, \mathcal{S}, P)$ is a probability space, we will use the notation

$$E_P(f) = \int_S f(x)P(dx).$$

If $P = P_X$ for some random variable $X : \Omega \rightarrow S$, we will use E_X rather than E_{P_X} .

1.4.7 Conditional expectations (general case)

We use (CE1) and (CE2) as a definition of conditional expectation in the general case. We assume that $(\mathcal{R}_X, \mathcal{S}_X)$ and $(\mathcal{R}_Y, \mathcal{S}_Y)$ are measurable spaces.

Definition 1.2. Assume that $\mathcal{R}_Y = \mathbb{R}^d$. Let $X : \Omega \rightarrow \mathcal{R}_X$ and $Y : \Omega \rightarrow \mathcal{R}_Y$ be two random variables with $\mathbb{E}(|Y|) < \infty$. The conditional expectation of Y given X is a random variable $Z : \Omega \rightarrow \mathcal{R}_Y$ such that:

- (i) There exists a measurable function $h : \mathcal{R}_X \rightarrow \mathbb{R}^d$ such that $Z = h \circ X$ almost surely.
- (ii) For any measurable function $g : \mathcal{R}_X \rightarrow [0, +\infty)$, one has

$$\mathbb{E}(Yg \circ X) = \mathbb{E}(Zg \circ X).$$

The variable Z is then denoted $\mathbb{E}(Y | X)$ and the function h in (i) is denoted $\mathbb{E}(Y | X = \cdot)$.

Importantly, random variables Z satisfying conditions (i) and (ii) always exist and are almost surely unique, in the sense that if another function Z' satisfies these conditions, then $Z = Z'$ with probability one. One obtains an equivalent definition if one restricts functions g in (ii) to indicators of measurable sets, yielding the condition that, if $A \subset \mathcal{R}_X$ is measurable,

$$\mathbb{E}(Y \mathbf{1}_{X \in B}) = \mathbb{E}(Z \mathbf{1}_{X \in B}).$$

With this general definition, we still have

$$\mathbb{E}(\mathbb{E}(Y | X)) = \mathbb{E}(Y)$$

and, for any function g defined on $\mathcal{R}_X$, $\mathbb{E}(Yg \circ X | X) = (g \circ X)\mathbb{E}(Y | X)$.

Conditional expectations share many of the properties of simple expectations. They are linear with respect to the Y variable. Moreover, if $Y \leq Y'$, both taking scalar values, then $\mathbb{E}(Y | X) \leq \mathbb{E}(Y' | X)$ almost surely. Jensen's inequality also holds: if $\gamma : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex and $\gamma \circ Y$ is integrable, then

$$\gamma \circ \mathbb{E}(Y | X) \leq \mathbb{E}(\gamma \circ Y | X).$$

We will discuss convex functions in chapter 3, but two important examples for this section are $\gamma(y) = |y|$ and $\gamma(y) = |y|^2$. The first one implies that $|\mathbb{E}(Y | X)| \leq \mathbb{E}(|Y| | X)$ and, taking expectations on both sides: $\mathbb{E}(|\mathbb{E}(Y | X)|) \leq \mathbb{E}(|Y|)$, the upper bound being finite by assumption. For the square norm, we find that, if Y is square integrable, then so is $\mathbb{E}(Y | X)$ and

$$\mathbb{E}(|\mathbb{E}(Y | X)|^2) \leq \mathbb{E}(|Y|^2).$$

If Y is square integrable, then this inequality shows that $\mathbb{E}(Y | X)$ minimizes $\mathbb{E}[|Y - Z|^2]$ among all square integrable functions $Z : \Omega \rightarrow \mathcal{R}_Y$ that satisfy (i). In other terms, the conditional expectation is the optimal least-square approximation of Y by a function of X . To see this, just write

$$\begin{aligned} \mathbb{E}(|Y - Z|^2 | X) &= \mathbb{E}(|Y|^2 | X) - 2\mathbb{E}(Y^T Z | X) + |Z|^2 \\ &= \mathbb{E}(|Y|^2 | X) - 2\mathbb{E}(Y | X)^T Z + |Z|^2 \\ &= \mathbb{E}(|Y|^2 | X) - |\mathbb{E}(Y | X)|^2 + |\mathbb{E}(Y | X) - Z|^2 \\ &= \mathbb{E}(|Y - \mathbb{E}(Y | X)|^2 | X) + |\mathbb{E}(Y | X) - Z|^2 \\ &\geq \mathbb{E}(|Y - \mathbb{E}(Y | X)|^2 | X) \end{aligned}$$

and taking expectations on both sides yields the desired result.

1.4.8 Conditional probabilities (general case)

If A is a measurable subset of $\mathcal{R}_Y$, then $\mathbf{1}_A$ is a random variable and its conditional expectation $\mathbb{E}(\mathbf{1}_A | X)$ (resp. $\mathbb{E}(\mathbf{1}_A | X = x)$) is denoted $\mathbb{P}(Y \in A | X)$ (resp. $\mathbb{P}(Y \in A | X = x)$), or $P_Y(A | X)$ (resp. $P_Y(A | X = x)$). Note that, for each A , these conditional probabilities is defined up to modifications on sets of probability zero, and it is not obvious that they can be defined for all A together (up to a modification on a common set of probability zero), since there is generally a non-countable number of sets A . This can be done, however, with some mild assumptions on the set $\mathcal{R}_Y$ and its σ -algebra (always satisfied in our discussions, see remark 1.1), ensuring that, for all $\omega \in \Omega$, $A \mapsto P_Y(A | X)(\omega)$ is a probability distribution on $\mathcal{R}_Y$ such that, for any measurable function $h : \mathcal{R}_Y \rightarrow \mathbb{R}$ such that $h \circ Y$ is integrable,

$$\mathbb{E}(h(Y) | X) = \int_{\mathcal{R}_Y} h(y) P_Y(dy | X).$$

Assume now that the sets $\mathcal{R}_X$ and $\mathcal{R}_Y$ are equipped with measures, say μ_X and μ_Y such that the joint distribution of (X, Y) is absolutely continuous with respect to $\mu_X \otimes \mu_Y$, so that there exists a function $\varphi : \mathcal{R}_X \times \mathcal{R}_Y \rightarrow \mathbb{R}$ (the p.d.f. of (X, Y) with respect to $\mu_X \otimes \mu_Y$) such that

$$\mathbb{P}(X \in A, Y \in B) = \int_{A \times B} \varphi(x, y) \mu_X(dx) \mu_Y(dy).$$

Then $P_Y(\cdot | X)$ is absolutely continuous with respect to μ_Y , with density given by the conditional p.d.f. of Y given X , namely,

$$\varphi(y | X) : \omega \mapsto \frac{\varphi(X(\omega), y)}{\int_{\mathcal{R}_Y} \varphi(X(\omega), y') \mu_Y(dy')} = \varphi(y | X = X(\omega)). \quad (1.13)$$

Note that

$$\mathbb{P} \left\{ \omega : \int_{\mathcal{R}_Y} \varphi(X(\omega), y') \mu_Y(dy') = 0 \right\} = 0$$

so that the conditional density can be defined arbitrarily when the numerator vanishes¹.

We retrieve here as special cases the definition of conditional probabilities and densities given in section 1.4.2. As an additional example, take $\mathcal{R}_X = \mathbb{R}^d$, μ_X being

¹Letting $\varphi_X(x) = \int_{\mathcal{R}_Y} \varphi(x, y') \mu_Y(dy')$, which is the marginal p.d.f. of X with respect to μ_X , we have

$$\mathbb{P}(\varphi_X(X) = 0) = \int_{\mathcal{R}_X} \mathbf{1}_{\varphi_X(x)=0} \varphi_X(x) \mu_X(dx) = 0.$$

Lebesgue's measure and assume that Y is discrete (so that $\mu_Y = \text{card}$), then

$$\varphi(y \mid X = X(\omega)) = \frac{\varphi(X(\omega), y)}{\sum_{y' \in \mathcal{R}_Y} \varphi(X(\omega), y')}.$$

Chapter 2

A Few Results in Matrix Analysis

This chapter collects a few results in linear algebra that will be useful in the rest of this book.

2.1 Notation and basic facts

We denote by $\mathcal{M}_{n,d}(\mathbb{R})$ the space of all $n \times d$ matrices with real coefficients¹. For a matrix $A \in \mathcal{M}_{n,d}(\mathbb{R})$ and integer $k \leq n$ and $l \leq d$, we let $A_{[k,l]} \in \mathcal{M}_{k,l}(\mathbb{R})$ denote the matrix A restricted to its first k rows and first l columns. The i, j entry of A will be denoted $A(i, j)$ or $A^{(ij)}$.

We assume that the reader is familiar with elementary matrix analysis, including, in particular the fact that symmetric matrices are diagonalizable in an orthonormal basis, i.e., if $A \in \mathcal{M}_{d,d}(\mathbb{R})$ is a symmetric matrix (whose space is denoted $\mathcal{S}_d$), there exists an orthogonal matrix $U \in \mathcal{O}_d$ (i.e., satisfying $U^T U = U U^T = \text{Id}_{\mathbb{R}^d}$) and a diagonal matrix $D \in \mathcal{M}_{d,d}(\mathbb{R})$ such that

$$A = U D U^T.$$

The identity $AU = UD$ then implies that the columns of U form an orthonormal basis of eigenvectors of A .

If $A \in \mathcal{S}_d^+$ is positive semi-definite (i.e., $u^T A u \geq 0$ for all $u \in \mathbb{R}^d$), the entries of D in the decomposition $A = U D U^T$ are non-negative, and one can define the matrix square root of A as $S = U D^{\odot 1/2} U^T$ where $D^{\odot 1/2}$ is the diagonal matrix formed taking the square roots of all coefficients of D . We will use the notation $S = A^{1/2}$. Note that $D^{1/2} = D^{\odot 1/2}$ if D is diagonal and positive semi-definite.

If $A \in \mathcal{S}_d^{++}$ is positive definite (i.e., A is positive semi-definite and $u^T A u = 0$ implies $u = 0$) and B is positive semi-definite, both being $d \times d$ matrices, the generalized

¹Unless mentioned otherwise, all matrices are assumed to be real.

eigenvalue problem associated with A and B consists in finding a diagonal matrix D and a matrix U such that $BU = AUD$ and $U^T AU = \text{Id}_{\mathbb{R}^d}$. Letting $\tilde{U} = A^{1/2}U$, the problem is equivalent to solving $A^{-1/2}BA^{-1/2}\tilde{U} = \tilde{U}D$ with $\tilde{U}^T \tilde{U} = \text{Id}_{\mathbb{R}^d}$, i.e., finding the eigenvalue decomposition of the symmetric positive-definite matrix $A^{-1/2}BA^{-1/2}$.

If $A \in \mathcal{M}_{n,d}(\mathbb{R})$, it can be decomposed as

$$A = UDV^T$$

where $U \in \mathcal{O}_n(\mathbb{R})$ and $V \in \mathcal{O}_d(\mathbb{R})$ are orthogonal matrices and $D \in \mathcal{M}_{n,d}(\mathbb{R})$ is diagonal (i.e., such that $D(i, j) = 0$ whenever $i \neq j$) with non-negative diagonal coefficients. These coefficients are called the singular values of A , and the procedure is called a *singular-value decomposition* (SVD) of A . An equivalent formulation is that there exist orthonormal bases $u_1, \dots, u_n$ of $\mathbb{R}^n$ and $v_1, \dots, v_d$ of $\mathbb{R}^d$ (forming the columns of U and V) such that

$$Av_i = \lambda_i u_i$$

for $i \leq \min(n, d)$, where $\lambda_1, \dots, \lambda_{\min(n, d)}$ are the singular values. Of course, if A is square and symmetric positive semi-definite, an eigenvalue decomposition of A is also a singular value decomposition (and the singular values coincide with the eigenvalues). More generally, if $A = UDV^T$, then $AA^T = UDD^T U^T$ and $A^T A = V D^T D V^T$ are eigenvalue decompositions of AA^T and $A^T A$. Singular values are uniquely defined, up to reordering. However, the matrices U and V are not unique up to column reordering in general. As a consequence, if A is a symmetric matrix and $A = UDV^T$ is a singular value decomposition of A , then the columns of both U and V provide eigenvectors of A^2 , hence of A , and $A = UDU^T = VDV^T$.

If $m = \min(n, d)$, then, forming the matrices $\tilde{U} = U_{[n, m]}$ (resp. $\tilde{V} = V_{[d, m]}$) by removing from U (resp. V) its last $n - m$ (resp. $d - m$) columns, and $\tilde{D} = D_{[m, m]}$ by removing from D its $n - m$ rows and $d - m$ columns, one has

$$A = \tilde{U} \tilde{D} \tilde{V}^T$$

with $\tilde{U}$, $\tilde{D}$ and $\tilde{V}$ having respectively size $n \times m$, $m \times m$ and $m \times d$, $\tilde{U}^T \tilde{U} = \tilde{V}^T \tilde{V} = \text{Id}_{\mathbb{R}^m}$ and $\tilde{D}$ diagonal with non-negative coefficients. This representation provides a *reduced SVD* of A and one can create a full SVD from a reduced one by completing the missing rows of $\tilde{U}$ and $\tilde{V}$ to form orthogonal matrices, and by adding the required number of zeros to $\tilde{D}$.

2.2 The trace inequality

We now describe Von Neumann's trace theorem. Its justification follows the proof given in Mirsky [141].

Theorem 2.1 (Von Neumann). *Let $A, B \in \mathcal{M}_{n,d}(\mathbb{R})$ have singular values $(\lambda_1, \dots, \lambda_m)$ and $(\mu_1, \dots, \mu_m)$, respectively, where $m = \min(n, d)$. Assume that these eigenvalues are listed in decreasing order so that $\lambda_1 \geq \dots \geq \lambda_m$ and $\mu_1 \geq \dots \geq \mu_m$. Then,*

$$\text{trace}(A^T B) \leq \sum_{i=1}^m \lambda_i \mu_i. \quad (2.1)$$

Moreover, if $\text{trace}(A^T B) = \sum_{i=1}^m \lambda_i \mu_i$, then there exist $n \times n$ and $d \times d$ orthogonal matrices U and V such that $U^T A V$ and $U^T B V$ are both diagonal, i.e., one can find SVDs of A and B in the same bases of $\mathbb{R}^n$ and $\mathbb{R}^d$.

Proof. We can assume without loss of generality that $d \leq n$ because, if the result holds for A and B , it also holds for A^T and B^T . Let $A = U_1 \Lambda V_1^T$ and $B = U_2 M V_2^T$ be the singular values decompositions of A and B (both Λ and M are $n \times d$ matrices). Then

$$\text{trace}(A^T B) = \text{trace}(V_1 \Lambda^T U_1^T U_2 M V_2) = \text{trace}(\Lambda^T U M V^T)$$

with $U = U_1^T U_2$ and $V = V_1^T V_2$. Let $u(i, j), 1 \leq i, j \leq n$ and $v(i, j), 1 \leq i, j \leq d$ be the coefficients of the orthogonal matrices U and V . Then

$$\text{trace}(\Lambda^T U M V^T) = \sum_{i,j=1}^d u(i, j) v(i, j) \lambda_i \mu_j \leq \frac{1}{2} \sum_{i,j=1}^d \lambda_i \mu_j u(i, j)^2 + \frac{1}{2} \sum_{i,j=1}^d \lambda_i \mu_j v(i, j)^2 \quad (2.2)$$

Let us consider the first sum in the upper-bound. Let $\xi_d = \lambda_d$ (resp. $\eta_d = \mu_d$) and $\xi_i = \lambda_i - \lambda_{i+1}$ (resp. $\eta_i = \mu_i - \mu_{i+1}$) for $i = 1, \dots, d-1$. Since singular values are non-increasing, we have $\xi_i, \eta_i \geq 0$ and

$$\lambda_i = \sum_{j=i}^d \xi_j, \quad \mu_i = \sum_{j=i}^d \eta_j$$

for $i = 1, \dots, d$. We have

$$\begin{aligned} \sum_{i,j=1}^d \lambda_i \mu_j u(i, j)^2 &= \sum_{i,j=1}^d \sum_{i'=i}^d \xi_{i'} \sum_{j'=j}^d \eta_{j'} u(i, j)^2 = \sum_{i',j'=1}^d \xi_{i'} \eta_{j'} \sum_{i=1}^{i'} \sum_{j=1}^{j'} u(i, j)^2 \\ &\leq \sum_{i',j'=1}^d \xi_{i'} \eta_{j'} \min(i', j') \end{aligned} \quad (2.3)$$

where we used the fact that U is orthogonal, which implies that $\sum_{j=1}^{j'} u(i, j)^2$ and $\sum_{i=1}^{i'} u(i, j)^2$ are both less than 1. Notice also that, when $u(i, j) = \delta_{ij}$ (i.e., $u(i, j) = 1$ if $i = j$ and zero otherwise), then

$$\sum_{i=1}^{i'} \sum_{j=1}^{j'} u(i, j)^2 = \min(i', j'),$$

so that the last inequality is an identity, and the chain of equalities leading to (2.3) implies

$$\sum_{i', j'=1}^d \xi_{i'} \eta_{j'} \min(i', j') = \sum_{i=1}^d \lambda_i \mu_j.$$

We therefore obtain (for any U), the fact that

$$\sum_{i, j=1}^d \lambda_i \mu_j u(i, j)^2 \leq \sum_{i=1}^d \lambda_i \mu_j.$$

The same identity obviously holds with v in place of u , and combining the two yields (2.1).

We now consider conditions for equality. Clearly, if one can find SVD decompositions of A and B with $U_1 = U_2$ and $V_1 = V_2$, then $U = \text{Id}_{\mathbb{R}^n}$, $V = \text{Id}_{\mathbb{R}^d}$ and (2.1) is an identity. We want to prove the converse statement.

For (2.1) to be an equality, we first need (2.2) to be an identity, which requires that $u(i, j) = v(i, j)$ as soon as $\lambda_i \mu_j > 0$. We also need an equality in (2.3), which requires

$$\sum_{i=1}^{i'} \sum_{j=1}^{j'} u(i, j)^2 = \min(i', j')$$

as soon as $\lambda_{i'} > \lambda_{i'+1}$ and $\mu_{j'} > \mu_{j'+1}$. The same identity must be true with $v(i, j)$ replacing $u(i, j)$

In view of this, denote by $i_1 < \dots < i_p$ (resp. $j_1 < \dots < j_q$) the indexes at which the singular values of A (resp. B) differ from their successors, with the convention $\lambda_{d+1} = \mu_{d+1} = 0$. Let, for $k = 1, \dots, p$ and $l = 1, \dots, q$

$$C(k, l) = \sum_{i=1}^{i_k} \sum_{j=1}^{j_l} u(i, j)^2.$$

Then, we must have $C(k, l) = \min(i_k, j_l)$ for all k, l and $u(i, j) = v(i, j)$ for $i = 1, \dots, i_p$ and $j = 1, \dots, j_q$.

If, for all $i, j \leq d$, we let $U_{[ij]}$ be the matrix formed by the first i rows and j columns of U , the condition $C_{kl} = \min(i_k, j_l)$ requires that $U_{[i_k j_l]} U_{[i_k j_l]}^T = \text{Id}_{\mathbb{R}^{i_k}}$ if $i_k \leq j_l$ and $U_{[i_k j_l]}^T U_{[i_k j_l]} = \text{Id}_{\mathbb{R}^{j_l}}$ if $j_l \leq i_k$. This shows that, if $i_k \leq j_l$, the rows of $U_{[i_k j_l]}$ form an orthonormal family, and necessarily, all elements $u(i, j)$ for $i \leq i_k$ and $j > j_l$ vanish. The symmetric situation holds if $j_l \leq i_k$.

Let $r_k = i_k - i_{k-1}$ and $s_l = j_l - j_{l-1}$ (with $i_0 = j_0 = 0$). We now consider possible changes in the SVDs of A and B . With our notation, the matrix Λ takes the form

$$\Lambda = \begin{pmatrix} \lambda_{i_1} \text{Id}_{\mathbb{R}^{r_1}} & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \lambda_{i_2} \text{Id}_{\mathbb{R}^{r_2}} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_{i_p} \text{Id}_{\mathbb{R}^{r_p}} & 0 & 0 & \cdots & 0 \\ 0 & & \cdots & & 0 & 0 & \cdots & 0 \\ \vdots & & & & \vdots & \vdots & & \vdots \\ 0 & & \cdots & & 0 & 0 & \cdots & 0 \end{pmatrix}$$

Let $W, \tilde{W}$ be $n \times n$ and $d \times d$ orthogonal matrices taking the form

$$W = \begin{pmatrix} W_1 & 0 & 0 & \cdots & 0 \\ 0 & W_2 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & W_p & 0 \\ 0 & & \cdots & & W_{p+1} \end{pmatrix}, \quad \tilde{W} = \begin{pmatrix} W_1 & 0 & 0 & \cdots & 0 \\ 0 & W_2 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & W_p & 0 \\ 0 & & \cdots & & \tilde{W}_{p+1} \end{pmatrix}$$

where $W_1, \dots, W_p$ are orthogonal with respective sizes $r_1, \dots, r_p$, W_{p+1} is orthogonal with size $n - i_p$ and $\tilde{W}_{p+1}$ is orthogonal with size $d - i_p$. Then we have

$$W D \tilde{W} = D$$

proving that U_1 can be replaced by $U_1 W$ provided that V_1 is replaced by $V_1 \tilde{W}$. Similar transformations can be made on U_2 and V_2 , with U_2 replaced by $U_2 Z$

and V_2 by $V_2\tilde{Z}$ with

$$Z = \begin{pmatrix} Z_1 & 0 & 0 & \cdots & 0 \\ 0 & Z_2 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & Z_q & 0 \\ 0 & & \cdots & & Z_{q+1} \end{pmatrix}, \quad \tilde{Z} = \begin{pmatrix} Z_1 & 0 & 0 & \cdots & 0 \\ 0 & Z_2 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & Z_q & 0 \\ 0 & & \cdots & & \tilde{Z}_{q+1} \end{pmatrix}$$

with a structure similar to W and $\tilde{W}$, replacing $r_1, \dots, r_p$ by $s_1, \dots, s_q$. As a consequence, $U = U_1^T U_2$ can be replaced by $W^T U Z$ and V by $\tilde{W}^T V \tilde{Z}$. To complete the proof, we need to show that, when (2.1) is an equality, these matrices can be chosen so that $W^T U Z = \text{Id}_{\mathbb{R}^n}$ and $\tilde{W}^T V \tilde{Z} = \text{Id}_{\mathbb{R}^d}$.

Let us consider a first step in this direction, assuming that $i_1 \leq j_1$ so that

$$U_{[i_1 j_1]} U_{[i_1 j_1]}^T = \text{Id}_{\mathbb{R}^{i_1}}.$$

Complete $U_{[i_1 j_1]}^T$ into a orthogonal matrix $Z_1 = [U_{[i_1 j_1]}^T, \tilde{U}]$. Build a matrix Z as above by taking $Z_2, \dots, Z_{q+1}$ equal to the identity. Then UZ has a first $i_1 \times i_1$ block equal to $\text{Id}_{\mathbb{R}^{i_1}}$, which implies that all coefficients on the right and below this block are zeros. If $j_1 \leq i_1$, a similar construction can be made on the other side, letting $W_1 = [U_{[i_1 j_1]} \tilde{U}]$ with the first $j_1 \times j_1$ block of the new matrix U equal to the identity. Note that, since $V_{[i_p j_q]} = U_{[i_p j_q]}$, the same result is obtained on V at the same time.

Pursuing this way (and skipping the formal induction argument, which is a bit tedious), we can progressively introduce identity blocks into U and V and transform them into new matrices (that we still denote by U and V) taking the form (letting $k = \min(i_p, j_q)$)

$$U = \begin{pmatrix} \text{Id}_{\mathbb{R}^k} & 0 \\ 0 & \tilde{U} \end{pmatrix} \text{ and } V = \begin{pmatrix} \text{Id}_{\mathbb{R}^k} & 0 \\ 0 & \tilde{V} \end{pmatrix}$$

If $k = i_p$ (resp. $k = j_q$), the final reduction can be obtained by choosing $W_{p+1} = \tilde{U}$ and $\tilde{W}_{p+1} = \tilde{V}$ (resp. $Z_{p+1} = \tilde{U}^T$ and $\tilde{Z}_{p+1} = \tilde{V}^T$), leading to SVDs for A and B with identical matrices $U_1 = U_2$ and $V_1 = V_2$. $\square$

Remark 2.2. Note that, since the singular values of $-A$ and of A coincide, theorem 2.1 implies

$$|\text{trace}(A^T B)| \leq \sum_{i=1}^m \lambda_i \mu_i. \quad (2.4)$$

for all matrices A and B , with equality if either A and B or $-A$ and B have an SVD

using the same bases. ♦

2.3 Applications

Let p and d be integers with $p \leq d$. Let $A \in \mathcal{S}_d(\mathbb{R})$, $B \in \mathcal{S}_p(\mathbb{R})$ be symmetric matrices. We consider the following optimization problem: maximize, over matrices $U \in \mathcal{M}_{d,p}(\mathbb{R})$ such that $U^T U = \text{Id}_{\mathbb{R}^p}$, the function

$$F(U) = \text{trace}(U^T A U B) = \text{trace}(A U B U^T).$$

We first note that the singular values of $U B U^T$, which is $d \times d$, are the same as the eigenvalues of B completed with zeros. Letting $\lambda_1 \geq \dots \geq \lambda_d$ be the eigenvalues of A and $\mu_1 \geq \dots \geq \mu_p$ those of B , we therefore have, from theorem 2.1,

$$F(U) \leq \sum_{i=1}^p \lambda_i \mu_i.$$

Introduce the eigenvalue decompositions of A and B in the form $A = V \Lambda V^T$ and $B = W M W^T$. For $F(U)$ to be equal to its upper-bound, we know that we must arrange $U B U^T$ to take the form

$$U B U^T = V \begin{pmatrix} M & 0 \\ 0 & 0 \end{pmatrix} V^T.$$

Use, as before, the notation $V_{[dp]}$ to denote the matrix formed with the p first columns of V . Take $U = V_{[dp]} W^T$, which satisfies $U^T U = \text{Id}_{\mathbb{R}^p}$. We then have

$$V_{[dp]} W^T B W V_{[dp]}^T = V_{[dp]} M V_{[dp]}^T = V \begin{pmatrix} M & 0 \\ 0 & 0 \end{pmatrix} V^T,$$

which shows that U is optimal. We summarize this discussion in the next theorem.

Theorem 2.3. *Let $A \in \mathcal{S}_d(\mathbb{R})$ and $B \in \mathcal{S}_p(\mathbb{R})$ be symmetric matrices, with $p \leq d$. Let eigenvalue decompositions of A and B be given by $A = V \Lambda V^T$ and $B = W M W^T$, where the diagonal elements of Λ (resp. M) are $\lambda_1 \geq \dots \geq \lambda_d$ (resp. $\mu_1 \geq \dots \geq \mu_p$).*

Define $F(U) = \text{trace}(A U B U^T)$, for $U \in \mathcal{M}_{d,p}(\mathbb{R})$. Then,

$$\max \{ F(U) : U^T U = \text{Id}_{\mathbb{R}^p} \} = \sum_{i=1}^p \lambda_i \mu_i.$$

This maximum is attained at

$$U = V_{[d,p]} W^T.$$

The following corollary applies theorem 2.3 with $B = \text{diag}(\mu_1, \dots, \mu_p)$.

Corollary 2.4. Let $A \in \mathcal{S}_d(\mathbb{R})$ be a symmetric matrix with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$. For $p \leq d$, let $\mu_1 \geq \dots \geq \mu_p > 0$ and define

$$F(e_1, \dots, e_p) = \sum_{i=1}^p \mu_i e_i^T A e_i.$$

Then, the maximum of F over all orthonormal families $e_1, \dots, e_p$ in $\mathbb{R}^d$ is $\sum_{i=1}^p \lambda_i \mu_i$ and is attained when $e_1, \dots, e_p$ are eigenvectors of A with eigenvalues $\lambda_1, \dots, \lambda_p$.

The minimum of F over all orthonormal families $e_1, \dots, e_p$ in $\mathbb{R}^d$ is $\sum_{i=1}^p \lambda_{d-i+1} \mu_i$ and is attained when $e_1, \dots, e_p$ are eigenvectors of A with eigenvalues $\lambda_d, \dots, \lambda_{d-p+1}$.

Proof. The statement about the maximum is just a special case of theorem 2.3, with $B = \text{diag}(\mu_1, \dots, \mu_p)$, noting that the i th diagonal element of $U^T A U$ is $e_i^T A e_i$ where e_i is the i th column of U .

The statement about the minimum is deduced by replacing A by $-A$. $\square$

Applying this corollary with $p = 1$, we retrieve the elementary result that $\lambda_1 = \max\{u^T A u : |u| = 1\}$ and $\lambda_d = \min\{u^T A u : |u| = 1\}$.

To complete this chapter, we quickly state and prove Rayleigh's theorem.

Theorem 2.5. Let $A \in \mathcal{M}_{d,d}(\mathbb{R})$ be a symmetric matrix with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$. Then

$$\lambda_k = \max_{V: \dim(V)=k} \min\{u^T A u, u \in V, |u| = 1\} = \min_{V: \dim(V)=d-k+1} \max\{u^T A u, u \in V, |u| = 1\}$$

where the min and max are taken over linear subspaces of $\mathbb{R}^d$.

Proof. Let $e_1, \dots, e_d$ be an orthonormal basis of eigenvectors of A associated with $\lambda_1, \dots, \lambda_d$. Let, for $k \leq d$, $W_{k,d} = \text{span}(e_k, \dots, e_d)$. Let V be a subspace of dimension k . Then $V \cap W_{k,d} \neq \emptyset$ (because the sum of the dimensions of these two spaces is $d + 1$). Taking u_0 with norm 1 in this intersection, we have

$$\min\{u^T A u, u \in V, |u| = 1\} \leq u_0^T A u_0 \leq \max\{u^T A u, u \in W_{k,d}, |u| = 1\} = \lambda_k,$$

where the last identity follows by considering the eigenvalues of A restricted to $W_{k,d}$. So, the maximum of the right-hand side is indeed less than λ_k , and it is attained for $V = W_{1,k}$. This proves the first identity, and the second one can be

obtained by applying the first one to $-A$. □

2.4 Some matrix norms

The *operator norm* of a matrix $A \in \mathcal{M}_{n,d}(\mathbb{R})$, is defined as

$$|A|_{\text{op}} = \max\{|Ax| : x \in \mathbb{R}^d, |x| = 1\}.$$

It is equal to the square root of the largest eigenvalue of $A^T A$, i.e., to the largest singular value of A .

The *Frobenius norm* of A is

$$|A|_F = \sqrt{\text{trace}(A^T A)} = \sqrt{\sum_{i,j=1}^d A(i,j)^2},$$

so that

$$|A|_F = \left(\sum_{k=1}^m \sigma_k^2 \right)^{1/2}$$

where $\sigma_1, \dots, \sigma_m$ are the singular values of A (and $m = \min(n, d)$).

The *nuclear norm* of A is defined by

$$|A|_* = \sum_{k=1}^d \sigma_k.$$

One can prove that this is a norm using an equivalent definition, provided by the following proposition.

Proposition 2.6. *Let A be an n by d matrix. Then*

$$|A|_* = \max \left\{ \text{trace}(UAV^T) : U \in \mathcal{M}_{n,n} \text{ and } U^T U = \text{Id}, V \in \mathcal{M}_{d,d} \text{ and } V^T V = \text{Id} \right\}.$$

Proof. The fact that $\text{trace}(UAV^T) \leq |A|_*$ for any U and V is a consequence of the trace inequality applied with $B = [\text{Id}, 0]$ or its transpose depending on whether $n \leq d$ or not. The upper-bound being attained when U and V are the matrices forming the singular value decomposition of A , the proof is complete. □

The fact that $|A|_*$ is a norm, for which the only non-trivial fact was the triangular inequality, now is an easy consequence of this proposition, because the maximum of the sum of two functions is always less than the sum of their maximums. More precisely, we have

$$\begin{aligned} |A+B|_* &= \max\{\text{trace}(UAV^T) + \text{trace}(UBV^T) : \\ &\quad U^T U = \text{Id}, V^T V = \text{Id}\} \\ &\leq \max\{\text{trace}(UAV^T) : U^T U = \text{Id}, V^T V = \text{Id}\} \\ &\quad + \max\{\text{trace}(UBV^T) : U^T U = \text{Id}, V^T V = \text{Id}\} \\ &= |A|_* + |B|_* \end{aligned}$$

The nuclear norm is also called the Ky Fan norm of order d . Ky Fan norms of order k (for $1 \leq k \leq d$) associate to a matrix A the quantity

$$|A|_{(k)} = \lambda_1 + \cdots + \lambda_k,$$

i.e., the sum of its k largest singular values. One has the following proposition.

Proposition 2.7. *The Ky Fan norms satisfy the triangular inequality.*

Proof. We prove this following the argument suggested in Bhatia [28]. For $A \in \mathcal{M}_{d,d}$, and $k = 1, \dots, d$, let $\text{trace}_{(k)}(A)$ be the sum of the k largest diagonal elements of A . Let, for a symmetric matrix A , $|A|'_{(k)}$ denote the sum of the k largest eigenvalues of A (it is equal to $|A|_{(k)}$ if A is positive definite, but can also include negative values).

Then, for any symmetric matrix $A \in \mathcal{S}_d$,

$$|A|'_{(k)} = \max\{\text{trace}_{(k)}(UAU^T) : U \in CO_d\}. \quad (2.5)$$

To show this, assume that V in $\mathcal{O}_d$ diagonalizes A , so that $D = VAV^T$ is a diagonal matrix. Assume, without loss of generality, that the coefficients $\lambda_j = D(j, j)$ are non-increasing. Fix $U \in \mathcal{O}_d$, let $B = UAU^T$ and $W = VU^T$ so that $D = WBW^T$, or $B = W^T D W$. Then, for any $j \leq d$,

$$B(j, j) = \sum_{i=1}^d W(i, j)^2 D(i, i).$$

Then, for any $1 \leq j_1 < \dots < j_k \leq d$

$$\begin{aligned}
\sum_{l=1}^k B(j_l, j_l) &= \sum_{i=1}^d D(i, i) \sum_{l=1}^k W(i, j_l)^2 \\
&= \sum_{i=1}^k D(i, i) + \sum_{i=1}^k D(i, i) \left(\sum_{l=1}^k W(i, j_l)^2 - 1 \right) + \sum_{i=k+1}^d D(i, i) \sum_{l=1}^k W(i, j_l)^2 \\
&= \sum_{i=1}^k D(i, i) + \sum_{i=1}^k (D(i, i) - D(k, k)) \left(\sum_{l=1}^k W(i, j_l)^2 - 1 \right) \\
&\quad + \sum_{i=k+1}^d (D(i, i) - D(k, k)) \sum_{l=1}^k W(i, j_l)^2 + D(k, k) \left(\sum_{i=1}^n \sum_{j=1}^k W(i, j_l)^2 - k \right).
\end{aligned}$$

Because W is orthogonal, we have $\sum_{l=1}^k W(i, j_l)^2 \leq 1$ and

$$\sum_{i=1}^n \sum_{j=1}^k W(i, j_l)^2 = k.$$

This shows that the terms after $\sum_{i=1}^k D(i, i)$ in the upper bound are negative or zero, so that

$$\sum_{l=1}^k B(j_l, j_l) \leq \sum_{i=1}^k D(i, i).$$

The maximum of the left-hand side is $\text{trace}_{(k)}(B)$. Noting that we get an equality when choosing $U = V$, the proof of (2.5) is complete.

Using the same argument as that made above for the nuclear norm, one deduces from this that

$$|A + B|'_{(k)} \leq |A|'_{(k)} + |B|'_{(k)}$$

for all $A, B \in \mathcal{S}_d$ and all $k = 1, \dots, d$.

Now, let $A \in \mathcal{M}_{n,d}$ and consider the symmetric matrix

$$\tilde{A} = \begin{pmatrix} 0 & A^T \\ A & 0 \end{pmatrix} \in \mathcal{S}_{n+d}.$$

Write a vector $u \in \mathbb{R}^{n+d}$ as $u = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ with $u_1 \in \mathbb{R}^d$ and $u_2 \in \mathbb{R}^n$. Then u is an eigenvector of $\tilde{A}$ for an eigenvalue λ if and only if $A^T u_2 = \lambda u_1$ and $A u_1 = \lambda u_2$,

which implies that $A^T A u_1 = \lambda^2 u_1$ and λ^2 is a singular value of A . Conversely, if μ is a nonzero singular value of A , associated with eigenvector u_1 , then $1/\sqrt{\mu}$ and $-1/\sqrt{\mu}$ are eigenvalues of $\tilde{A}$, associated with eigenvectors $\begin{pmatrix} u_1 \\ \pm A u_1 / \sqrt{\mu} \end{pmatrix}$. It follows from this that $|A|_{(k)} = |\tilde{A}|'_{(k)}$ for $k \leq \min(n, d)$ and therefore satisfies the triangle inequality. $\square$

We refer to [28] for more examples of matrix norms, including, in particular those provided by taking p th powers in Ky Fan's norms, defining

$$|A|_{(k,p)} = (\lambda_1^p + \cdots + \lambda_k^p)^{1/p}.$$

2.5 Low-rank approximation

Singular-values decompositions provide optimal approximations of general matrices by low-rank ones in terms of the Frobenius norm. Take $p \leq \min(n, d)$. Optimal rank- p approximations of an (n, d) matrix X in this sense are matrices Z^* that minimize $\|X - Z\|_F^2$ among all matrices Z of rank p . Rank- p matrices are described in the following proposition.

Proposition 2.8. *An (n, d) matrix Z has rank p if and only if it can be written in the form $Z = A W^T$ where A is (n, p) , and W is (d, p) with $W^T W = \text{Id}_{\mathbb{R}^p}$, i.e., $W = [e_1, \dots, e_p]$ where the columns form an orthonormal family of $\mathbb{R}^d$.*

Proof. The “if” part is obvious and we focus on the “only if” part. Assume that Z has rank p . Take $W = [e_1, \dots, e_p]$, where $(e_1, \dots, e_p)$ is an orthonormal family in $\text{Null}(Z)^\perp$. Letting $e_{p+1}, \dots, e_d$ denote an orthonormal basis of $\text{Null}(Z)$, we have $\sum_{i=1}^d e_i e_i^T = \text{Id}_{\mathbb{R}^d}$ and

$$Z = Z \sum_{i=1}^d e_i e_i^T = Z \sum_{i=1}^p e_i e_i^T = Z W W^T$$

so that one can take $A = Z W$. $\square$

Taking $Z = A W^T$ as in the proposition, we have

$$\|X - A W^T\|_F^2 = \text{trace}(X X^T) - 2\text{trace}(X W A^T) + \text{trace}(A A^T).$$

With fixed W , the optimal A is given by $A = X W$, so that W must minimize

$$\text{trace}(X X^T) - \text{trace}(X W W^T X^T) = \text{trace}(X^T X) - \text{trace}(W W^T X^T X),$$

i.e., maximize $\text{trace}(WW^T X^T X)$. Let $m = \min(n, d)$ and $\lambda_1 \geq \dots \geq \lambda_m$ the singular values of X . The singular values of $X^T X$ and $\lambda_1^2 \geq \dots \geq \lambda_m^2$, possibly completed with zeros if $d > m$. Those of WW^T and 1, with multiplicity p and 0, with multiplicity $d - p$. The trace inequality shows that the maximum value of $\text{trace}(WW^T X^T X)$ is $\lambda_1^2 + \dots + \lambda_p^2$, and is attained when both $X^T X$ and WW^T have singular values decompositions in the same bases, which implies, since these are symmetric matrices, that they have a shared basis of eigenvectors. This shows that taking $W = e_1 e_1^T + \dots + e_p e_p^T$ provides a solution, where $e_1, \dots, e_p$ are eigenvectors of $X^T X$ associated with $\lambda_1^2, \dots, \lambda_p^2$, or, equivalently, the first column vectors of V in the singular values decomposition $X = UDV^T$. With our notation, this is $W = V_{[d,p]}$ and

$$A = XW = UDV^T V_{[d,p]} = UD_{[n,p]}.$$

We summarize this in the next proposition.

Proposition 2.9. *Let X be an (n, d) matrix and $X = UDV^T$ be a singular value decomposition of X . Then an optimal rank- p approximation of Z is provided by the matrix*

$$Z = XV_{[d,p]} V_{[d,p]}^T = UD_{[n,p]} V_{[d,p]}^T.$$

Chapter 3

Introduction to Optimization

This chapter summarizes some fundamental concepts in optimization that will be used later in the book. The reader is referred to textbooks, such as Beck [22], Eiselt et al. [70], Nocedal and Wright [150], Boyd et al. [39] and many others for proofs and deeper results.

3.1 Basic Terminology

1. If I is a subset of $\mathbb{R}$, a lower bound of I is an element $u \in [-\infty, +\infty]$ such that $u \leq x$ for all $x \in I$. Among these lower bounds, there exists a largest element, denoted $\inf I \in [-\infty, +\infty]$, called the infimum of I (by convention, the infimum of an empty set is $+\infty$). Similarly, one defines the supremum of I , denoted $\sup I$, as the smallest upper bound of I (and the supremum of an empty set is $-\infty$). Every set in $\mathbb{R}$ has an infimum and a supremum, but these numbers do not necessarily belong to I . When they do, they are respectively called minimal and maximal elements of I , and are denoted $\min I$ and $\max I$. So, the statement “ $u = \min I$ ” means $u \in I$ and $u \leq v$ for all $v \in I$.

2. If $F : \Omega \rightarrow \mathbb{R}$ is a real-valued function defined on a subset $\Omega \subset \mathbb{R}^d$, the infimum of F over Ω is defined by

$$\inf_{\Omega} F = \inf\{F(x) : x \in \Omega\}$$

and its supremum is

$$\sup_{\Omega} F = \sup\{F(x) : x \in \Omega\}.$$

As seen above both numbers are well defined, and can take infinite values. One says that $x \in \Omega$ is a (global) minimizer (resp. maximizer) of F if $F(y) \geq F(x)$ (resp. $F(y) \leq F(x)$) for all $y \in \Omega$. One also says that F reaches its minimum (resp. maximum), or is minimized (resp. maximized) at x . Equivalently, x is a minimizer (resp. maximizer) of F if and only if $x \in \Omega$ and

$$F(x) = \min\{F(y) : y \in \Omega\} \text{ (resp. } \max\{F(y) : y \in \Omega\}).$$

In such cases, one also writes $F(x) = \min_{\Omega} F$ or $F(x) = \max_{\Omega} F$. In particular, the notation $u = \min_{\Omega} F$ indicates that $u = \inf_{\Omega} F$ and that there exists an x in Ω such that $F(x) = u$ (i.e., that the infimum of F over Ω is realized at some $x \in \Omega$). Note that the infimum of a function always exists, but not necessarily its minimum. Also note that minimizers, when they exist, are not necessarily unique. We will denote by $\operatorname{argmin}_{\Omega} F$ (resp. $\operatorname{argmax}_{\Omega} F$) the (possibly empty) set of minimizers (resp. maximizers) of F .

3. One says that x is a local minimizer (resp. maximizer) of F on Ω if there exists an open ball $B \subset \mathbb{R}^d$ such that $x \in B$ and $F(x) = \min_{\Omega \cap B} F$ (resp. $F(x) = \max_{\Omega \cap B} F$).

4. An optimization problem consists in finding a minimizer or maximizer of an “objective function” F . Focusing from now on on minimization problems (statements for maximization problems are symmetric), we will always implicitly assume that a minimizer exists. The following provides some general assumptions on F and Ω that ensure this fact.

The sublevel sets of F in Ω are denoted $[F \leq u]_{\Omega}$ (or simply $[F \leq u]$ when $\Omega = \mathbb{R}^d$) for $u \in [-\infty, +\infty]$ with

$$[F \leq u]_{\Omega} = \{x \in \Omega : F(x) \leq u\}.$$

Note that

$$\operatorname{argmin}_{\Omega} F = \bigcap_{u > \inf F} [F \leq u]_{\Omega}.$$

A typical requirement for F is that its sublevel sets are closed in $\mathbb{R}^d$, which means that, if a sequence (x_n) in Ω satisfies, for some $u \in \mathbb{R}$, $F(x_n) \leq u$ for all n and converges to a limit x , then $x \in \Omega$ and $F(x) \leq u$. If this is true, one says that F is *lower semi-continuous*, or l.s.c., on Ω . If, in addition to being closed, the sublevel sets of F are bounded (at least for u small enough—larger than $\inf F$), then $\operatorname{argmin}_{\Omega} F$ is an intersection of nested compact sets, and is therefore not empty (so that the optimization problem has at least one solution).

5. Different assumptions on F and Ω lead to different types of minimization problems, with specific underlying theory and algorithms.

1. If F is C^1 or smoother and $\Omega = \mathbb{R}^d$, one speaks of an unconstrained smooth optimization problem.

2. For constrained problems, Ω is often specified by a finite number of inequalities, i.e.,

$$\Omega = \{x \in \mathbb{R}^d : \gamma_i(x) \leq 0, i = 1, \dots, q\}.$$

If F and all functions $\gamma_1, \dots, \gamma_q$ are C^1 one speaks of smooth constrained problems.

3. If Ω is a convex set (i.e., $x, y \in \Omega \Rightarrow [x, y] \in \Omega$, where $[x, y]$ is the closed line segment connecting x and y) and F is a convex function (i.e., $F((1-t)x + ty) \leq (1-t)F(x) + tF(y)$ for all $x, y \in \Omega$), one speaks of a convex optimization problem.

4. Non-smooth problems are often considered in data science, and lead to interesting algorithms and solutions.

5. When both F and $\gamma_1, \dots, \gamma_q$ are affine functions, one speaks of a linear programming problem (or a linear program). (An affine function is a mapping $x \mapsto b^T x + \beta$, $b \in \mathbb{R}^d$, $\beta \in \mathbb{R}$.)

If F is quadratic ($F(x) = \frac{1}{2}x^T A x - b^T x$), and all γ_i 's are affine, one speaks of a quadratic programming problem.

6. Finally, some machine learning problems are specified over discrete or finite sets Ω (for example $\mathbb{Z}^d$, or $\{0, 1\}^d$), leading to combinatorial optimization problems.

3.2 Unconstrained Optimization Problems

3.2.1 Conditions for optimality (general case)

Consider a function $F : \Omega \rightarrow \mathbb{R}$ where Ω is an *open* subset of $\mathbb{R}^d$. We first discuss the unconstrained optimization problem of finding

$$x^* \in \underset{\Omega}{\operatorname{argmin}} F. \quad (3.1)$$

The following result summarizes (non-identical) necessary and sufficient conditions that are applicable to such a solution.

Theorem 3.1. Necessary conditions. Assume that F is differentiable over Ω , and that x^* is a local minimum of F . Then $\nabla F(x^*) = 0$.

If F is C^2 , then, in addition, $\nabla^2 F(x^*)$ must be positive semidefinite.

Sufficient conditions. Assume that $F \in C^2(\Omega)$. If $x^* \in \Omega$ is such that $\nabla F(x^*) = 0$ and $\nabla^2 F(x^*)$ is positive definite, then x^* is a local minimum of F .

Proof. Necessary conditions: Since Ω is open, it contains an open ball centered at x^* , with radius ϵ_0 and therefore all segments $[x^*, x^* + \epsilon h]$ for all $\epsilon \in [0, \epsilon_0]$ and all unit norm vectors h . Since x^* is a local minimum, we can choose ϵ_0 so that $F(x^* + \epsilon h) \geq F(x^*)$ for all h with $|h| = 1$.

Using Taylor formula, we get (for $\epsilon \in [0, \epsilon_0]$, $|h| = 1$)

$$0 \leq F(x^* + \epsilon h) - f(x^*) = \epsilon \int_0^1 dF(x^* + t\epsilon h) h dt.$$

If $dF(x^*)h \neq 0$ for some h , then, for small enough ϵ , $dF(x^* + t\epsilon h)h$ cannot change sign for $t \in [0, 1]$ and therefore $\int_0^1 dF(x^* + t\epsilon h)h dt$ has the same sign as $dF(x^*)(h)$ which must therefore be positive. But the same argument can be made with h

replaced by $-h$, implying that $dF(x^*)(-h) = -dF(x^*)h$ is also positive, and this gives a contradiction. We therefore have $dF(x^*)(h) = 0$ for all h , i.e., $\nabla F(x^*) = 0$.

Assume that F is C^2 . Then, making a second-order Taylor expansion, one gets

$$0 \leq F(x^* + \epsilon h) - F(x^*) = \epsilon^2 \int_0^1 (1-t) d^2F(x^* + t\epsilon h)(h, h) dt.$$

The same argument as above shows that, if $d^2F(x^*)(h, h) \neq 0$, then it must be positive. This shows that $d^2F(x^*)(h, h) \geq 0$ for all h and $d^2F(x^*)$ (or its associated matrix $\nabla^2 F(x^*)$) is positive semidefinite.

Now, assume that F is C^2 and $\nabla^2 F(x^*)$ positive definite. One still has

$$F(x^* + \epsilon h) - F(x^*) = \epsilon^2 \int_0^1 (1-t) d^2F(x^* + t\epsilon h)(h, h) dt$$

If $\nabla^2 F(x^*) > 0$, then $\nabla^2 F(x^* + t\epsilon h) > 0$ for small enough ϵ , showing the r.h.s. of the identity is positive for $h \neq 0$, and that $F(x^* + \epsilon h) > F(x^*)$. $\square$

Because maximizing F is the same as minimizing $-F$, necessary (resp. sufficient) conditions for optimality in maximization problems are immediately deduced from the above: it suffices to replace positive semidefinite (resp. positive definite) by negative semidefinite (resp. negative definite).

3.2.2 Convex sets and functions

Definition 3.2. One says that a set $\Omega \subset \mathbb{R}^d$ is convex if and only if, for all $x, y \in \Omega$, the closed segment $[x, y]$ also belongs to Ω .

A function $F : \mathbb{R}^d \rightarrow (-\infty, +\infty]$ is convex if, for all $\lambda \in [0, 1]$ and all $x, y \in \mathbb{R}^d$, one has

$$F((1-\lambda)x + \lambda y) \leq (1-\lambda)F(x) + \lambda F(y). \quad (3.2)$$

If, whenever the lower bound is not infinite, the inequality above is strict for $\lambda \in (0, 1)$, one says that F is strictly convex.

Note that, with our definition, convex functions can take the value $+\infty$ but not $-\infty$. In order for the upper-bound to make sense when F takes infinite values, one makes the following convention: $a + (+\infty) = +\infty$ for any $a \in (-\infty, +\infty]$; $\lambda \cdot (+\infty) = +\infty$ for any $\lambda > 0$; $0 \cdot (+\infty)$ is not defined but $0 \cdot (+\infty) + (+\infty) = +\infty$.